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The Strategic Effects of Trademark Protection

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We study the effects of trademark protection on firms' profits and strategy using the 1996 Federal Trademark Dilution Act, which granted additional legal protection to selected trademarks. We find that the FTDA raised treated firms' operating profits and was followed by a spike in trademark lawsuits and lower entry and exit in affected product markets. Treated firms reduced R&D spending, produced fewer patents and new products, and recalled a higher number of unsafe products. Our results suggest that stronger trademark protection negatively affected innovation and product quality. (*JEL* D22, K2, L43, O31, O34, O38)

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Introduction

As the U.S. economy shifts toward service- and technology-based industries, firm value is increasingly accounted for by intangible capital, such as intellectual property (IP). This paper examines a basic class of IP assets—trademarks—and presents evidence on the effects of trademark protection on firms' profits and strategy. Trademarks grant the holder a monopoly over a particular brand. The efficiency rationale for trademark protection is that it incentivizes firms to invest in product quality and development. On the other hand, stronger trademark protection inevitably insulates incumbents from

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competition. Whether the quality-incentive or monopoly-rent effect dominates is an empirical question, with significant policy implications.

To study the causal effects of varying trademark protection we exploit the Federal Trademark Dilution Act (FTDA) of 1996, which granted additional legal protection to “famous" trademarks until its key provision was nullified in 2003 by a U.S. Supreme Court decision. We find that the FTDA raised treated firms’ operating return on assets by an average of 1.7 percentage points (pp), equal to 12% of their average pretreatment profits. The FTDA was followed by a sharp increase in trademark lawsuits under the new provision and by reduced entry and turnover in affected goods and service classes, consistent with our hypothesis that the FTDA raised the expected cost of entry into affected product markets. Treatment effects were strongest in firms that faced moderate entry threat ex ante, consistent with strategic entry deterrence as a mechanism.

We examine the FTDA’s effects on product quality, innovation and product market strategy. In theory, trademark protection incentivizes firms to produce high-quality products and prevents a race to the bottom (Klein and Leffler 1981; Landes and Posner 1987). Alternatively, according to theories dating back to Chamberlin (1933), trademark protection insulates incumbents from competition, in which case stronger protection may lead to more exploitative behavior. We find that firms granted stronger trademark protection had increased frequency and dollar value of recalls of unsafe products and were less likely to launch a recall voluntarily.

Theory also predicts that trademark protection affects innovation. First, some inventions cannot be patented or are better protected via secrecy instead of disclosure.[1] Second, patents expire while trademarks do not: pharmaceutical firms often continue to sell the branded drug at a premium after their patent expires. Third, trademark protection is a determinant of market power, which is a primary incentive for innovation (Arrow 1962). We find that treated firms reduced R&D spending, patenting activity, and new product introductions, suggesting that stronger trademark protection led to lower competition and less innovation as hypothesized in Aghion and Schankerman (2004). We also find that treated firms altered their product market strategy by introducing brand-extending products in new product categories. At the same time, these firms created fewer new products in their legacy product categories. Taken together, our results suggest that firms responded to stronger trademark protection by pursuing a more exploitative and less innovative product market strategy, at the same time extending their brands into all-new product markets.

The main challenge for our research design is that the FTDA neglected to define the term “famous," and the fame of each trademark was then decided in court on a case-by-case basis. Thus, no objective rule governs whether any given trademark qualified as “famous." Two aspects of our research design help

[1] Coca-Cola, Kentucky Fried Chicken, and WD-40 are three examples of brands built on a secret formula.

mitigate this problem. First, the key question for the FTDA's effects on firm behavior is whether the firm and its competitors believed a trademark would qualify. Guided by the legal literature, we explore three independent approaches to assigning treatment status as of 1995 and find similar results for all three. Second, in our research design, mistakes in assigning treatment status represent errors-in-variables and attenuate the estimates toward zero as long as the errors are not correlated with changes in the outcome variables (Lewbel 2007). With this caveat, our results can be seen as a lower bound on the law's true effects.

Our setting lets us identify the effects of the treatment on the treated, that is, the effects of granting additional antidilution protection to incumbent firms. Thus, our results do not speak directly to the effects of protection against infringement, which is the fundamental right associated with a trademark. However, our results are informative about changes to trademark policy at the margin, in an environment (the United States) of strong IP enforcement. Interestingly, our results on product quality are consistent with those of Qian (2008) in a very different setting: Qian finds that name-brand Chinese shoe companies responded to *weaker* protection against counterfeits by producing *higher* quality products.

A causal interpretation of our findings rests on the assumption of parallel trends: absent the FTDA, changes in the profits and strategies of firms that held famous trademarks would have been similar to those of control firms. The main threat to validity is unobserved changes at the firm level that coincided with the FTDA and with famous trademark holdings. We perform a battery of tests that provide additional depth and lend confidence to a causal interpretation.

First, the timing of the estimated effect is sharply limited to the treatment window. Inspection of the year-by-year differences between treated and control firms shows similar pretreatment trends and a positive break in treated firms' profits after the FTDA took effect. A nearly identical trademark protection law was withdrawn from a prior bill that took effect in 1989 and estimates around this placebo event show no effects. Second, our estimates are stable across a range of alternative specifications. Our main estimates include both firm fixed effects, which remove observed and unobserved non-time-varying heterogeneity between firms, and industry-by-year fixed effects, which expel arbitrary industry booms and busts. Our estimates are also robust to including individual firm-level pretrends. Third, we find that the FTDA's effects systematically varied with proxies for the ex ante importance of trademark protection and were strongest in firms that faced moderate entry threat ex ante, both of which are consistent with our proposed mechanism.

A large literature highlights the importance of brands to consumers and firms.[2] For example, Bronnenberg et al. (2015) document an average 29% premium of name-brand goods over identical generics across a broad sample

[2] See, for example, Bronnenberg, Dhar, and Dubé (2009), Bronnenberg, Dube, and Gentzkow (2012), Larkin (2013), Vitorino (2013), Belo, Lin, and Vitorino (2014).

of consumer-packaged goods. Trademarks are the primary legal instrument that allows firms to differentiate their name-brand goods from the competition, yet there has been little research on the effects of trademark protection on firm value and strategy, due in part to a lack of comprehensive data. Our results and our new mapping from Compustat to the recently released USPTO trademark data files help to fill that gap.

To our knowledge, our paper is the first to demonstrate a causal link between trademarks and firms' profits, value, and strategy. Previous studies have found that a firm's stock of trademarks is positively correlated with cash flow, Tobin's q, return on assets, and stock returns, and negatively correlated with cash flow volatility.[3] Previous studies also find positive correlations between trademark activity and measures of innovation (Mendonca, Pereira, and Godinho 2004; Faurel et al. 2016) and venture capital valuation (Block et al. 2014).

Further, we document empirical evidence consistent with Sutton (1991) and Shaked and Sutton (1987), who argue that firms invest in differentiating their products and creating endogenous barriers to entry. Hoberg and Phillips (2010) apply textual analysis to firms' 10-K product descriptions and find that both R&D and advertising are associated with subsequent differentiation from competitors. Consistent with endogenous differentiation, we find that industries more affected by the FTDA had less "creative destruction" as measured by firm entry and exit and new product creation. We also document nonlinearities in treated firms' response depending on their ex ante entry threat, consistent with strategic entry deterrence.[4]

Our paper makes three main contributions. First, we analyze the comprehensive USPTO data on trademarks and connect it to firm data in Compustat. Although trademarks are well established as one of the three basic forms of IP in the United States along with patents and copyrights, research into trademark protection has previously been limited due to a lack of data. Second, we demonstrate and measure a causal link between trademark protection and firms' profits and strategy. Third, we provide new evidence that strengthening trademark protection was followed by lower product quality and innovation by treated firms. These results add to recent studies of intellectual property rights that suggest patent protection might actually hinder innovation (Williams 2013; Moser 2013).

1. Background on U.S. Trademarks

1.1 Trademark registration and renewal

The USPTO defines a trademark as "a word, phrase, symbol, design, color, smell, sound, or combination thereof that identifies and distinguishes the goods

3 See, for example, Kerin and Sethuraman (1998), Madden, Fehle, and Fournier (2006), Krasnikov, Mishra, and Orozco (2009), Sandner and Block (2011).

4 See, for example, Dafny (2005), Ellison and Ellison (2011), Frésard and Valta (2016), and Cookson(2017a, 2017b).

and services of one party from those of others."[5] Examples of trademarks include the word Pepsi, the McDonald's Golden Arches symbol, and NBC's musical notes G, E, C played on chimes. A trademark's distinctiveness can change over time; the once-strong trademarks Dry Ice, Escalator and Aspirin eventually became genericized" and ineligible for renewal (Graham et al. 2013).

The main statute of modern trademark law, the Lanham Act of 1946, defines infringement as "use of an identical or similar mark that would cause confusion as to the source of goods or services." The symbol ™ signifies an unregistered trademark, which is protected from infringement by state-level common law within the geographic area in which the mark is used. The symbol ® signifies that a trademark is officially registered with the USPTO. Registration extends protection against infringement to the national level, provides prima facie evidence of ownership, the power to file actions in Federal court to obtain injunctions and recover damages, and the mark is listed with the U.S. Customs and Border Protection Service, which interdicts the import of counterfeit goods.

The registrant specifies one or more goods-and-services classes, which define the scope of trademark protection. The current U.S. trademark classification system contains 52 goods codes and 8 services codes. Registration costs a small fee (as of 2016, \$325 per class) and requires that the applicant demonstrate "use in commerce," that is, proof that the mark is currently used to identify a good or service that they sell.[6] Each class in which a trademark is registered is subject to the use-in-commerce requirement at registration and renewal. The use-in-commerce requirement is important because it means that registered trademarks reflect products and services that firms were verified to produce and sell (Graham et al. 2013).

Unlike patents and copyrights, which expire after a limited time, trademarks may be renewed indefinitely. Prior to 1989 trademarks were renewed every 20 years; subsequently, they are renewed every 10 years. Proof of use in commerce is required again 6 years after registration and upon renewal.[7] The requirement to continue to make and sell a nontrivial quantity of the trademarked product, as well as the responsibility of legally protecting trademarks, imposes a significant maintenance cost that prevents squatting on unused trademarks. The trademark data are consistent with this assertion, as 53% of registered trademarks are allowed to expire after 6 years, implying that more than one-half of registered

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[5] The "Trademark Basics" section of the USPTO Web site (http://www.uspto.gov/trademarks-getting-started/trademark-basics) offers more details.

[6] Applications can be filed on an "intent-to-use" basis, but the applicant must demonstrate use in commerce before the registration can be completed.

[7] In the sixth year following trademark registration, the owner must file a Section 8 declaration that the trademark is currently in use. Subsequently, every 10 years following registration another Section 8 declaration and a Section 9 renewal form must be submitted. See the USPTO registration time line at (https://www.uspto.gov/trademark).

trademarks do not have sufficient value to justify the cost of maintaining them (Marco and Myers 2015).

1.2 Trademark dilution

In the decades after the Lanham Act of 1946 the concept of trademark dilution began to play an increasing role in litigation (Derenberg 1956). Dilution is a much broader concept than infringement and posits that a trademark has broader importance than simply allowing the buyer to identify the seller. “The value is in the ’aura’ of the mark... and the feelings it evokes from consumers about anything associated with that brand name" (Welkowitz 2012). Dilution is legally defined as any action “weakening... a famous mark’s ability to identify and distinguish goods or services *regardless of competition in the marketplace*” (italics added).[8]

Prior to 1996, protection or remedy from trademark dilution was granted at the state level in cases of proven dilution only (Oswald 1999). The resultant patchwork of state-level statutes and precedents prompted calls for federal antidilution legislation that would better protect nationally recognized brands (Duffy and Nagel 1997; Welkowitz 2012). The first attempt was in the Trademark Law Revision Act (TLRA), which was enacted in 1988. The antidilution provisions of the TLRA in draft form were almost identical to those of the later FTDA, but freedom of speech concerns led to the removal of that section from the TLRA shortly before its passage (Denicola 1997). We employ the TLRA as a placebo test.

The Federal Trademark Dilution Act (FTDA), which was signed into law on January 16, 1996, for the first time granted federal protection to U.S. trademarks against dilution. The FTDA was a major expansion of trademark rights in that a trademark holder was no longer required to prove actual infringement but only to convince a judge of the likelihood of dilution in order to obtain an injunction (Kim 2001; Bickley 2011). As an example of the law’s sweeping effects, in *Nabisco, Inc., v. PF Brands, Ltd. (1999)* Pepperidge Farms, the producer of Goldfish crackers, obtained an injunction to stop Nabisco’s intended selling of crackers based on cartoon characters some of which would be fish shaped.[9] In the case filings Nabisco estimated they had already spent $3.4 million on the project for licensing and development.[10]

A key limitation was that the FTDA specified that only “famous” trademarks qualified for federal protection against likely dilution. However, the FTDA did not define the term “famous.” What constituted a famous mark was subsequently

[8] Quoted from the INTA Trademark Dilution Factsheet (http://www.inta.org/TrademarkBasics/FactSheets/Pages/TrademarkDilution.aspx).

[9] *Nabisco, Inc., v. PF Brands, Inc., 50 F. Supp. 2d 188, 1999.*

[10] The Internet Appendix describes a number of additional court cases post-1995 that illustrate the expansive protection the FTDA granted to trademark holders.

interpreted on a case-by-case basis and was the subject of much debate (Duffy and Nagel 1997; Becker 2000; Dollinger 2001).

Figure 1 plots the number of court cases by year in the LexisNexis database that included a claim of trademark dilution. We split cases by whether they contained state dilution claims alone or a federal dilution claim. As state-level dilution claims are much more limited in scope and importance, we consider a case with both federal and state dilution claims to be a federal dilution case. In 1996, the year of the FTDA's passage, the rate of trademark dilution lawsuits nearly doubled from its 1995 rate due entirely to claims under the new federal statute. Filings of dilution lawsuits continued to rise over the next several years, to more than triple the pre-FTDA rate. However, there was no significant change in new trademark registrations (see the Internet Appendix). These observations are consistent with the view from the legal literature that the FTDA provided established trademarks—and only established trademarks—with significant new protections (Zando-Dennis 2004; Jacobs 2004).

There were two major legal developments post-FTDA. In March 2003 the U.S. Supreme Court ruled in *Moseley v. V. Secret Catalogue, Inc.* (2003) that a successful dilution claim required proof of actual economic damages, effectively nullifying the FTDA's key provision. The *Moseley* ruling was seen as a rebuke to the FTDA's overly broad legal standard for dilution (Pulliam 2003). Consistent with this view, Figure 1 illustrates how the number of federal dilution claims fell sharply starting in 2003.[11]

The Trademark Dilution Revision Act (TDRA) of 2006, explicitly drafted by legislators as a response to *Moseley*, restored the ability of famous trademark holders to sue on the basis of likely dilution without proving damages. However, the TDRA added provisions that altered and reduced the scope of protection relative to the FTDA and was perceived as failing to restore the pre-*Moseley* status quo (Cendali and Schriefer 2006; Beebe 2007).

2. Data and Methods

2.1 Trademarks data

Data on U.S. trademarks directly come from the USPTO.[12] Altogether 4,201,205 trademarks were registered at the USPTO between 1870 and 2012. We match trademarks in the year they were registered to Compustat firm-years using a variety of methods to ensure a comprehensive match. First, we collect all names of trademark grantees and run a fuzzy match to firm names from CRSP and Compustat and to parent and subsidiary firm names from CapitalIQ. We procure the name-to-gvkey associations from the NBER patent database, because many trademark assignees are also patent assignees

[11] The Internet Appendix provides detailed examples of lawsuits with dilution claims and their outcomes.

[12] Graham et al. (2013) describe the data in detail.

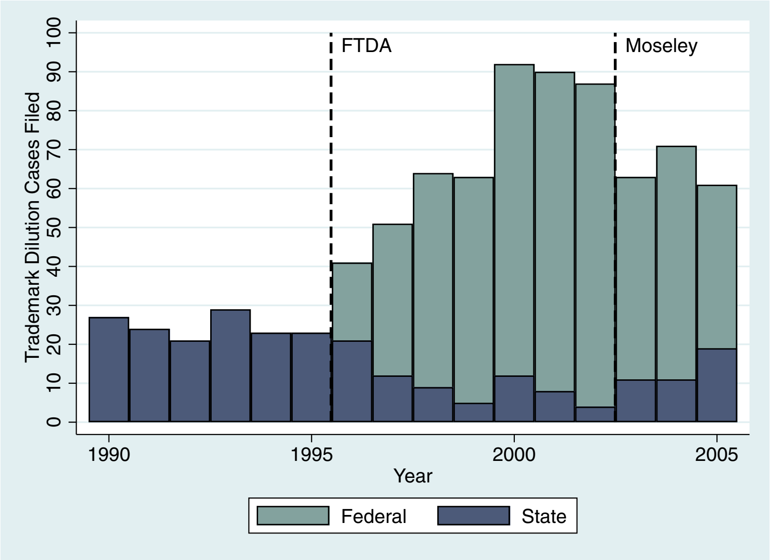

Figure 1
The figure displays the number of trademark dilution cases filed in U.S. district courts by year, 1990–2005. The total number of cases by year is divided by the type of dilution claim: the area on top represents cases with federal dilution claims, and the area below represents cases with state dilution claims.

(Hall, Jaffe, and Trajtenberg 2001). We double-check and supplement these automated matches with manual matching. In total we match 521,997 trademarks to 14,703 Compustat firms. The oldest trademark we match to a firm is Octagon Soap, which was registered to Colgate Inc. in 1888 and expired in 1998.

We record the USPTO goods-and-services classes for each trademark. We compute which classes each firm is active in, and was ever previously active in, each year. Based on the firm-level active classes we compute whether each new trademark a firm registers is in an old class, defined as a class in which the firm previously held at least one trademark, or a new class, defined as a class in which the firm never previously held a trademark. Also, in each class we compute entry and exit of firms (private and public) each year based on the disambiguated assignee names in the data.

We also record whether each trademark was subsequently renewed, and whether the text of the trademark (the brand name) was a new brand or an extension of an existing brand; for example, "Coke Zero" is an extension of Coca-Cola's existing brand "Coke."

2.2 Product recalls

We collected all 10,725 product recalls announced by the Consumer Product Safety Commission (CPSC) and National Highway Traffic Safety

Administration (NHTSA) from 1982 to 2005. Using a combination of fuzzy and manual matching, we matched 3,167 recalls to Compustat firm-years. We also record the quantity of the item(s) recalled and their average price where stated. For the NHTSA recalls we also record whether the recall was voluntarily initiated by the manufacturer or by the NHTSA's Office of Defects Investigation (ODI).

2.3 Patent and new product announcements data

We use the USPTO patent data from Kogan et al. (2017) as our primary measure of innovation.[13] These data cover 1,801,879 patents with nonmissing market values from 1926 to 2010. Another measure of product innovation is new product announcements gathered from newspapers or magazines (Chaney, Devinney, and Winer 1991; Mishra and Bhabra 2001). To capture product innovations that were not covered by patents, we gather new product announcements from the *New York Times* digital archives. Specifically, we scrape articles that mention a Compustat firm and the phrase "new product." For each article or advertisement meeting these criteria we add one to the count of new product announcements for the firm in the year the article was published. We repeat this process for all firms in our panel for the years 1982–2005.

2.4 Firm data and summary statistics

Firm accounting data come from Compustat. We retain U.S. firms from 1982 to 2005 with book assets of at least $1 million and with nonmissing and nonnegative revenue and market value of equity. These screens yield a sample of 139,142 firm-years for 16,389 firms. All accounting variables are winsorized at the 1% and 99% levels. The Internet Appendix defines all variables. We define industries at the NAICS 4-digit level.

We determine a firm's stock of trademarks directly using registration, renewal and expiry dates for every USPTO trademark. Table 1 presents summary statistics of the sample, comparing firm-years that did and did not hold at least one trademark. 50.8% of firms in the sample held at least one trademark. On average, trademark holders were larger, more profitable and older, had similar Tobin's q, market-to-book ratio, investment, leverage, and cash holdings, and spent more on advertising and R&D compared to firms that never held a trademark.

2.5 Research design

To study the causal impact of trademark protection on firms' profits and strategy, we exploit variation in the legal protection of a subset of trademarks over time. Two features of the 1996 Federal Trademark Dilution Act (FTDA) are important

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[13] https://kelley.iu.edu/nstoffma/.

Table 1
Summary statistics

	Non trademark holders			Trademark holders		
	Mean	*Median*	*SD*	*Mean*	*Median*	*SD*
logSales	3.61	3.55	1.99	5.11	5.09	2.2
logAssets	4.33	4.15	2.13	5.25	5.04	2.25
ROA	0	0.0503	0.278	0.0561	0.108	0.24
q	8.7	0.738	194	1.91	0.623	27.1
Market/Book	3.18	1.55	5.36	3.04	1.81	4.39
BookLeverage	0.275	0.201	0.621	0.235	0.183	0.313
Capex/Assets	0.0859	0.0381	0.225	0.0702	0.046	0.107
Cash/Assets	0.159	0.0631	0.218	0.171	0.0804	0.21
Ad/Assets	0.00987	0	0.0529	0.0173	0	0.0582
R&D/Assets	0.0319	0	0.0941	0.0519	0.00293	0.102
TrademarkStock	0	0	0	31.7	6	96.5
Age	12.4	9	14.8	19.1	16	16
NewPatents	0.206	0	2.15	7.99	0	61.7
Recalls	0.0012	0	0.0487	0.0459	0	0.7165
Number of firm-years	72,094			67,048		

The table presents summary statistics of the main sample, which is an unbalanced panel of U.S. firms from 1982 to 2005. The table is divided between firm-years that held at least one trademark and firm-years without trademarks.

for our purposes. First, until the passage of the FTDA, federal law protected against direct infringement and not the much broader concept of likely dilution. Thus, the FTDA represented a significant strengthening of trademark protection relative to previous years. Second, the FTDA explicitly limited protection against likely dilution to "famous" trademarks, although it did not define the term (Becker 2000; Dollinger 2001; Bickley 2011).

In our main specification we classify a trademark as plausibly affected by the FTDA if it was registered in 1974 or earlier and was still active on January 16, 1996. Thus, as of the FTDA's effective date, a plausibly famous trademark was renewed at least once and had been active in commerce for at least the previous 21 years. We define the registration cutoff at the end of 1974 to allow 1 year for notifications and processing of renewals and expiries. To check whether endogenous renewals are a threat to validity, in unreported checks we move the registration cutoff to 1973 or 1972 and require that the trademark was renewed by the end of 1994 or 1993; our results are identical.

Because the FTDA itself did not specify objective criteria for famousness, any definition is likely to produce both false negatives and false positives. Both types of error represent errors-in-variables in our research design and thus attenuate our estimates downward toward finding no difference between treated and control firms, as long these errors are not correlated with *changes* in outcome variables around the FTDA.[14] The Internet Appendix explores

[14] We examine two alternative classifications—Google Books and *Financial World* magazine—as well as intersections and unions between the proxy measures. We also use a matching estimator to match treated and control firms as of 1995. The results are similar in every case.

alternative classifications and matched estimates, and the results are similar in every case.

As it is common to take out multiple trademarks across different subproducts and goods and service classes (i.e., Ford Taurus, Ford F-150, Ford F-250, Ford hats, Ford key chains, Ford financing programs), the number of famous trademarks that a firm holds need not correspond with the size of the market or importance of the brand. Therefore, we classify firms as treated or control based on whether they held zero (control group) or one or more (treated group) plausibly famous trademarks as of 1995, the last pretreatment year in the sample. We classify 729 firms in 170 NAICS4 industries as treated and 6,271 firms as controls.

On average treated firms were older and larger than control firms and held more trademarks. However, there is still substantial overlap across firm size, age, and trademark holdings, because firms pursued different branding strategies over time. Consider the grocery store chains Winn-Dixie and American Stores. As of 1995, American Stores was 55% larger by sales, 30 years older, and held 66 live trademarks to Winn-Dixie's 37. But because American Stores went through a series of rebranding initiatives in the 1980s, all its trademarks were 10 years or less in age compared to Winn-Dixie's core brand, which dates to 1955. Hence, we code Winn-Dixie as a treated firm and American Stores as a control firm. The Internet Appendix discusses covariate balance and common support in more detail and shows that our results are robust when we match the treated and control groups *ex ante*.

3. Effects of the FTDA on Firms' Profits

Table 2 presents difference-in-differences estimates of the effect of the Federal Trademark Dilution Act on treated firms' operating profits.[15]

The specification is

$$ROA_{it} = \beta\,\text{FamousTM1995}_i \times \text{PostFTDA}_t + \gamma X_{it} + \phi_i + \tau_t + \lambda_{jt} + \epsilon_{it} \qquad (1)$$

where firms are indexed by i, industries are indexed by j, and ϕ_i, τ_t , λ_{jt} are firm, year, and industry-by-year fixed effects. $FamousTM1995_i$ is a dummy variable that equals 1 if the firm held one or more famous trademarks in 1995. $PostFTDA_t$ is a dummy variable that equals 0 if the year is in 1989-1995 and 1 if the year is in 1996–2002. X_{it} is a set of firm-year controls: q_{it}, $Ad/Sales_{it}$, $R\&D/Sales$, Age_{it}, $logTrademarkStock_{it}$. The set includes direct controls for firm age and the size of the firm's trademark portfolio, as

[15] Our measure of profitability is return on assets (ROA), defined as operating income before depreciation divided by book assets. This is a measure of operating profits that excludes more discretionary items, such as depreciation, interest, and R&D expenses. The results are very similar when we use alternative measures, such as operating profit after depreciation or net income over assets instead.

Table 2
Effects on operating profits

	(1) *ROA*	(2) *ROA*	(3) *ROA*	(4) *ROA*	(5) *ROA*	(6) *ROA*	(7) *ROA*
$FamousTM1995_i \times$ $PostFTDA_t$	0.020*** (5.7)	0.017*** (4.2)	0.017*** (4.0)			0.015*** (3.5)	0.015*** (3.5)
$FamousTM2002_i \times$ $Post2002_t$				−0.010** (−2.2)	−0.011** (−1.7)	−0.011*** (−2.3)	−0.009* (−1.8)
Observations	84,842	84,227	72,493	62,684	52,771	133,721	116,473
Adjusted R^2	.648	.658	.672	.686	.697	.632	.645
Firm-year Controls	No	No	Yes	No	Yes	No	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	No	No	No	No	No	No
Industry x Year FEs	No	Yes	Yes	Yes	Yes	Yes	Yes

The table presents difference-in-differences estimates of the effects of the 1996 Federal Trademark Dilution Act, and the subsequent nullification of its main provision in *Moseley v. V. Secret Catalogue, Inc.*, in 2003, on firms' operating profits (ROA). *FamousTM1995* and *FamousTM2002* are dummy variables that equal one if the firm held at least one plausibly famous trademark at the end of 1995 and 2002, respectively. The sample runs from 1989-2002 for columns 1-3, 1996-2005 for columns 4 and 5, and 1989-2005 for columns 6 and 7. The firm-year controls are *q*, *Ad/Sales*, *R&D/Sales*, *Age*, *logTrademarkStock*. Industry-by-year fixed effects are at the 4-digit NAICS level. Standard errors are robust and clustered by firm.

well as Tobin's q, ad spending, and R&D spending. Our preferred specification does not include the control variables, because if those variables are affected by the treatment themselves, then including them produces biased estimates (Angrist and Pischke 2008).

Table 2, Column 1, uses firm and year fixed effects and finds a 2.0 pp post-FTDA increase in ROA for treated firms relative to control firms. Column 2, our preferred specification, uses firm and industry-by-year fixed effects where industries are defined at the 4-digit NAICS level. This specification adjusts firms' profits so that the remaining variation is within each industry-year. The estimated treatment effect is +1.7 pp. Column 3 adds firm-year controls and the estimate is unchanged. The effect is economically significant, representing 12% of treated firms' average pretreatment ROA, which was 14.3%.

On March 4, 2003, the U.S. Supreme Court ruled in *Moseley v. Victoria's Secret* that a federal claim of trademark dilution required proof of actual economic damages, which was considered a de facto nullification of the protection granted by the FTDA (Pulliam 2003). Table 2, Columns 4 and 5, present difference-in-differences estimates around the *Moseley* ruling:

$$ROA_{it} = \beta\, \text{Famous2002}_i \times \text{Post2002}_t + \phi_i + \tau_t + \lambda_{jt} + \epsilon_{it} \qquad (2)$$

Here, the pretreatment period of 1996–2002 is when the FTDA was in effect and the post-treatment period of 2003–2005 is after its key provision was nullified. If our hypothesis is correct, then we expect treatment effects that are opposite in sign and similar in magnitude to those of the FTDA. Indeed, Columns 4 and 5 show that the *Moseley* ruling was accompanied by a change in profits among treated firms of −1.1 pp, opposite in sign and similar in magnitude to our main estimates around the FTDA's introduction, and this estimate is again robust to adding firm-year controls.

In Columns 6 and 7, we estimate the FTDA's effects using a switching research design. Doing so pools (1) and (2) across all firm-years from 1989 to 2005, so that the effects of the FTDA's passage in 1996 and the *Moseley* decision in 2003 are estimated jointly. The coefficients are again consistent with strengthening trademark protection in 1996 and removing it in 2003.

Figure 2 plots the average ROA for the treated and control groups in each year after firm and industry-by-year fixed effects. Panel A shows that the trends of two groups in the pretreatment period are similar and there is no significant difference between the groups in any pretreatment year. Consistent with our hypothesis and with the results in Table 2, there is a clear positive trend break in ROA for treated firms when the FTDA took effect in 1996, and no significant change in trend for control firms. Again consistent with our hypothesis and main estimates, the difference in average ROA between treated and control groups largely disappears in the 3 years after the *Moseley* decision.

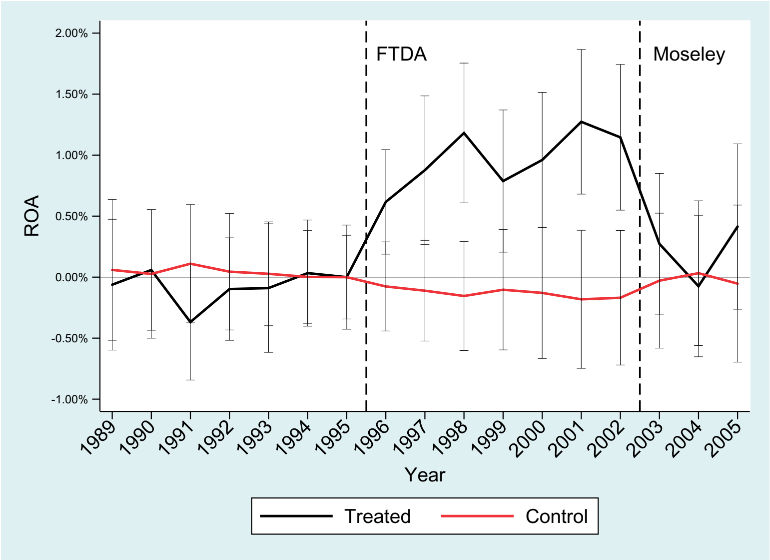
We also conduct placebo tests centered on the Trademark Law Revision Act (TLRA) of 1998. In draft form, the TLRA contained an antidilution provision that was almost identical to that of the later FTDA, but the provision was removed from the TLRA shortly before its passage (Gilson 1993; Denicola 1997). Figure 2, panel B, shows that treated and control firms trended similarly both before and after the passage of the TLRA. This suggests that reverse causality or some omitted variables (e.g., anticipated increases in incumbent firms' profits or market demand causing those firms to advocate for antidilution protection laws) are unlikely to explain our results.

3.1 Robustness checks

We perform additional checks and analysis, detailed in the Internet Appendix, that we summarize here. First, we hypothesize that trademark protection should be more valuable in industries that are more focused on sales and marketing and in industries that relied more on trademark protection ex ante. Indeed, we find that the effects of the FTDA were concentrated in industries that were ad intensive, selling intensive, and trademark intensive. Further, the effects of the FTDA were stronger for firms with more geographically dispersed operations circa 1995, which plausibly corresponds with firms' need for *federal* dilution protection. We also find that equity markets recognized the value of the profits that followed the FTDA, as treated firms' book-to-market ratios rose after the passage of the FTDA and fell following the *Moseley* decision.

As seen in Table 1, trademark holders were generally larger, more profitable and older, had similar average Tobin's q, market-to-book ratio, investment, leverage and cash holdings, and spent more on advertising and R&D compared to firms that never held a trademark. We enforce common support and covariate balance in the pretreatment period using coarsened exact matching (Iacus, King, and Porro 2012) and find similar results. We also find that our results

A.Effects of FTDA and Moseley



B. Placebo test around TLRA

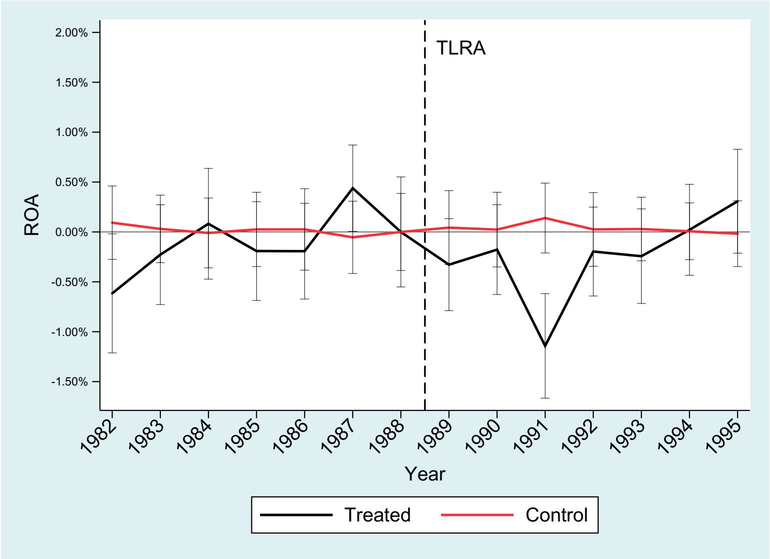


Figure 2
The figure displays the yearly average residual ROA for the treated and control firm groups after panel regressions with firm and industry-by-year fixed effects. The figure also shows 95% confidence intervals for each group mean in each year. To ease trend comparison, we align the two series at zero in the last pretreatment year. In panel A the dashed lines represent the passage of the Federal Trademark Dilution Act and the nullification of its main provision in *Moseley v. V. Secret Catalogue, Inc.* In panel B the dashed line represents the passage of the Trademark Law Revision Act.

are robust to a simple nonparametric comparison of pre-post changes by firm as recommended by Bertrand, Duflo, and Mullainathan (2004). The Internet Appendix details these and other tests.

4. Effects of the FTDA on Firm Strategy

4.1 Product recalls

The central argument in favor of trademark protection is that trademarks incentivize firms to produce high quality products and prevent a race to the bottom in product quality. We investigate this central pillar of the FTDA's effects on firm strategy using a broad and objective measure: recalls of unsafe products.[16]

The Consumer Product Safety Commission (CPSC) conducts safety testing and enforces recalls of most consumer products in the United States; the National Highway Traffic Safety Administration (NHTSA) conducts safety testing and enforces recalls of automotive products in the United States. Unsafe products are detected through several channels. First, firms are required by law to conduct appropriate internal safety tests and report defects as soon as they are discovered. Second, the agencies conduct proactive testing in their own labs. Third, the agencies operate hotlines and Web sites that allow consumers to report unsafe products. We collected data on all product recalls announced by the CPSC and NHTSA between 1982 and 2005.

Table 3 presents difference-in-differences estimates of the FTDA's effects on product recalls. Column 1, using a logit model, shows that post-FTDA treated firms were almost twice as likely (+81%) to have a product recalled in any given year. Column 2 shows that the results are similar using a linear probability model: The estimated treatment effect is +1.5 pp compared to a base rate of 0.9 pp, and Columns 3 and 4 show similar results using the number of product recalls, either in a generalized negative binomial model or in logs. Columns 5 and 6 show that the number of units recalled rose 17% and the dollar value of products recalled rose 31% for treated firms relative to control firms in the same industry-year.[17]

These results suggest that the FTDA's added trademark protection resulted in lower product quality among treated firms. An alternative explanation is that the rise in recall frequency reflects greater caution by treated firms; for example, if the increased value of their brands led treated firms to conduct more careful testing and proactively recall unsafe products. In Table 3, Column 7, the dependent variable $Voluntary$ is the fraction of recalls that were voluntarily

[16] Most measures of product quality used in the literature are limited to specific industries and settings (Fang 2005; Qian 2008; Matsa 2010; Phillips and Sertsios 2013) and are not well suited to a broad cross-section of firms and industries.

[17] A doubling of the recall rate appears large compared to the smaller responses observed for ROA and other variables. However, because recalls are rare events that occur in only 1%–2% of firm-years, the increase represents a doubling from a low baseline. Moreover, announcement returns suggest that the direct costs of product recalls are small, less than 1% of firm value on average (see the Internet Appendix).

Table 3
Effects on product quality

	(1) *Had Recall*	(2) *Had Recall*	(3) *Recalls*	(4) *log (Recalls)*	(5) *log (Units)*	(6) *log ($Value)*	(7) *Voluntary*
$FamousTM1995_i$ $\times PostFTDA_t$	0.81***	0.015***	0.16	0.015***	0.17***	0.31***	−0.051
	(3.4)	(3.4)	(0.9)	(3.0)	(3.5)	(4.8)	(−0.5)
$log Sales$	1.05***	0.0014***	0.96***	0.0016***	0.017***	0.015**	−0.098
	(14.8)	(2.9)	(19.5)	(3.2)	(3.2)	(2.5)	(−0.7)
Model	Logit	OLS	Neg. bin.	OLS	OLS	OLS	OLS
Observations	88,326	86,578	88,326	86,578	86,578	86,294	199
Adjusted R^2		.448		.675	.459	.316	.261
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	No	Yes	No	No	No	No
Industry × Year FEs	No	Yes	No	Yes	Yes	Yes	Yes

The table presents difference-in-differences estimates of the effects of the 1996 Federal Trademark Dilution Act on firms' product recalls. *FamousTM1995* is a dummy variable that equals one if the firm held at least one plausibly famous trademark at the end of 1995. *HadRecall* is a dummy variable that equals one if the firm announced at least one product recall that year. *Recalls* is the number of recalls by firm-year. *Units* is the number of product units recalled, and *$Value* is the total dollar value of products recalled, where reported. *Voluntary* is the fraction of recalls that were initiated by the firm. Industry-by-year fixed effects are at the 4-digit NAICS level. Standard errors are robust and clustered by firm.

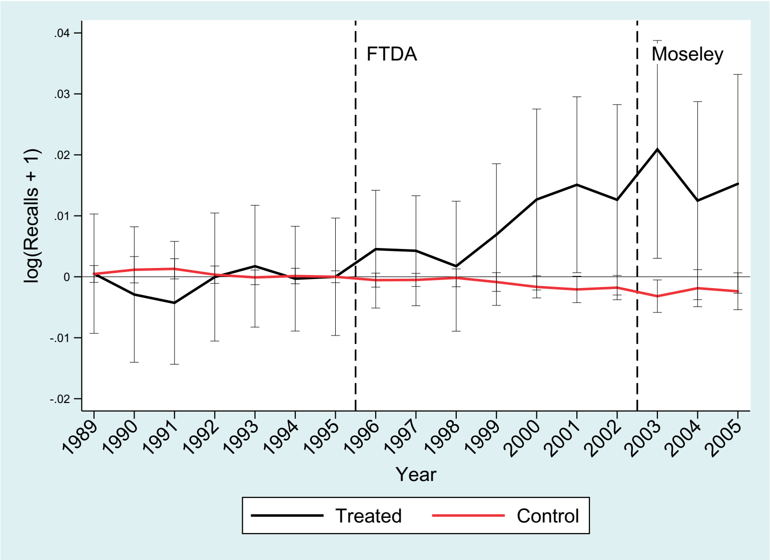
initiated by the firm itself. The estimated treatment effect is not statistically significant, and the point estimate is negative (-5.1 pp); that is, treated firms were slightly less likely to initiate a recall voluntarily.

Figure 3 shows graphs of the year-by-year coefficients for the number of recalled products surrounding the TLRA, FTDA, and Moseley decision. Panel A shows that the trends for the treatment and control groups in the pretreatment period prior to the FTDA are similar. Unlike the observed effects on firms' profits in Figure 2, the effect of the FTDA on product quality is more gradually realized. The apparent lag in the treatment effect is plausibly due to the time needed for an unsafe product to be produced, sold, detected, and recalled.[18] Consistent with the hypothesis that the increase in product recalls was an effect of the FTDA, panel B shows no treatment effect around the placebo event of the TLRA.

In sum, across a range of specifications, post-FTDA treated firms declared more product recalls covering more units with a higher dollar value and were less likely to do so voluntarily. These results suggest that treated firms lowered their product quality in the post-FTDA period and are broadly consistent with the quiet life (Bertrand and Mullainathan 2003) and trademarks-foster-monopoly hypotheses (Chamberlin 1933). In the absence of an incentive for higher quality, stronger trademark protection will lead firms to raise prices and cut costs as they face fewer competitive threats. If product quality is costly to

[18] Outcomes, such as recalls that react with a lag, limit our estimates to the post-FTDA period, prior to the *Moseley* reversal. The post-*Moseley* period from 2003 to 2006 is not enough time to observe the treatment effects, especially because Congress immediately began drafting a law that would reverse the *Moseley* ruling.

A. Effects of FTDA and Moseley



B. Placebo test around TLRA

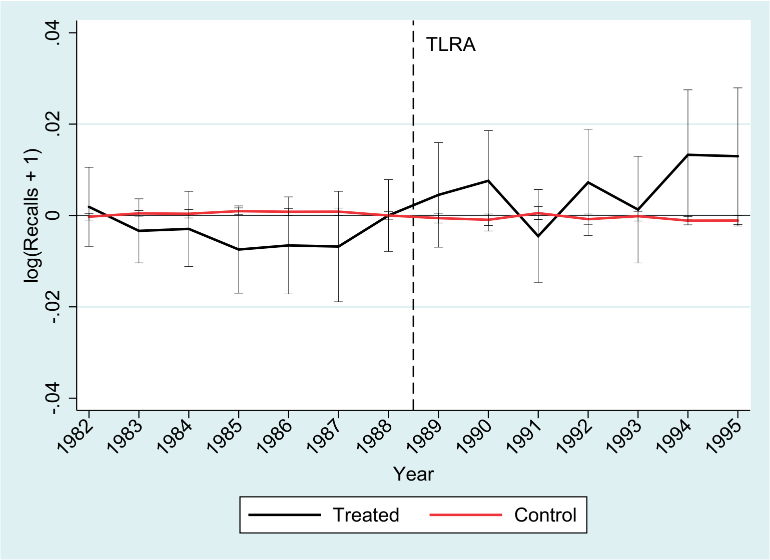


Figure 3
The figure displays the yearly average residual of product recalls for the treated and control firm groups after panel regressions with firm and industry-by-year fixed effects. The figure also shows 95% confidence intervals for each group mean in each year. To ease trend comparison, we align the two series at zero in the last pretreatment year. In panel A the dashed lines represent the passage of the Federal Trademark Dilution Act and the nullification of its main provision in *Moseley v. V. Secret Catalogue, Inc.* In panel B the dashed line represents the passage of the Trademark Law Revision Act.

maintain, via more stringent internal testing and lower production yields, then firms are likely to accept a cut on these margins as well.

Our results are also potentially consistent with the quality-incentive hypothesis (e.g., Shapiro 1983), because we measure the effects of dilution protection on treated firms, which were incumbents that owned famous brands. The quality-incentive hypothesis suggests that firms will produce high-quality products initially to build a reputation. Once reputation is established, an incumbent will trade off the costs of higher quality (hiring more inspectors, lower production yields) against the business lost by producing a lower quality product. Strengthening trademark protection for incumbents might increase their temptation to produce lower quality at a higher markup. Our results are also notably consistent with those of Qian (2008) in a very different setting. Qian finds that weaker protection against counterfeits in China led brand-name shoe firms to produce higher quality goods; we find that stronger trademark protection in the United States led treated firms to produce lower quality goods.

4.2 Innovation

Next we investigate the relationship between trademark protection and innovation. A large theoretical literature predicts that the incentive to innovate depends on firms' market power, specifically the difference between expected preinnovation and postinnovation rents (Arrow 1962; Aghion and Howitt 1992; Aghion et al. 2009). Both the quiet life hypothesis (Bertrand and Mullainathan 2003) and the Chamberlin (1933) rent extraction hypothesis suggest that stronger trademark protection leads to more managerial slack or more preinnovation rents, and thus to less innovation. Alternatively, the higher profits we document might have relaxed financial constraints and been at least partly plowed back into R&D.

Table 4 investigates several measures of firms' innovative activities. Column 1 shows that treated firms reduced R&D spending over assets by 0.32 pp following the FTDA. This is a significant change relative to their pretreatment mean of 3.4%. Column 2 shows that treated firms took out 9% fewer patents post-FTDA. When we weight patents by the adjusted citations they subsequently received, the reduction is greater at −14% (Column 3). However, the estimated dollar value of granted patents for treated firms (from Kogan et al. 2017) actually increased post-treatment by 13% (Column 4). The asymmetry of the citation- and dollar-weighted estimates is striking, and consistent with the FTDA leading treated firms to pursue a less exploratory and more exploitative innovation strategy.

Apart from R&D and patents, a common measure of product-level innovation is announcements of major new product lines gathered from newspapers or trade magazines (Chaney, Devinney, and Winer 1991; Mishra and Bhabra 2001). Table 4 Column 5 shows that treated firms were 13% less likely to announce a major new product in the post-treatment period, with the number of new

Table 4
Effects on firm innovation

	(1) *R&D/ Assets*	(2) *log (Patents)*	(3) *log (Citations)*	(4) *log (Value)*	(5) *Product*	(6) *log (Products)*
$FamousTM1995_i$ $\times PostFTDA_t$	−0.0032*** (−2.6)	−0.093*** (−3.7)	−0.14*** (−4.6)	0.13*** (3.2)	−0.13*** (−12.2)	−0.24*** (−11.4)
Observations	86,578	86,578	86,578	86,578	86,578	86,578
Adjusted R^2	.768	.837	.801	.836	.527	.633
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Year FEs	Yes	Yes	Yes	Yes	Yes	Yes

The table presents difference-in-differences estimates of the effects of the 1996 Federal Trademark Dilution Act on firms' innovation activity. *FamousTM1995* is a dummy variable that equals one if the firm held at least one plausibly famous trademark at the end of 1995. *R&D/Assets* is a firms' R&D expenditure over book assets. *Patents* is the number of granted patents the firm applied for in year t. *Citations* is the total weighted citations to the granted patents, from Kogan et al. (2017). *Value* is the estimated dollar value of the granted patents, from Kogan et al. (2017). *Product* is a dummy variable that equals one if the firm announced at least one new product that year in the press. *Products* is the number of new product announcements. Industry-by-year fixed effects are at the 4-digit NAICS level. Standard errors are robust and clustered by firm.

products announced showing a 24% decrease (Column 6). Taken together, these results appear consistent with both the quiet life and trademarks-foster-monopoly hypotheses, or a model in which the FTDA discouraged innovation by raising preinnovation rents.

Figure 4 shows graphs of the year by year coefficients for the main innovation outcomes, R&D spending and patents. For brevity the graphs span 1982–2005 to include the TLRA, FTDA, and the *Moseley* decision. Panel A shows that treated firms' R&D spending was unchanged around the TLRA placebo event, fell after the passage of the FTDA, and reversed to pretreatment levels following the *Moseley* decision. Panel B shows that new patent grants were unchanged around the TLRA placebo event, fell after the passage of the FTDA, and leveled off following the *Moseley* decision. These year-by-year coefficient estimates are consistent with the estimates in Table 4 and with our hypotheses that the following the FTDA, treated firms moved from an innovative toward an exploitative product market strategy.

4.3 Mechanism: Increased barriers to entry

Stronger trademark protection did not alter firms' production technology or consumers' demand for their products. Our hypothesis is that the FTDA raised the expected entry costs into treated product markets, because firms granted antidilution protection could sue for damages or injunctive relief over a much broader set of competitors' actions.

We use the trademark data to directly examine how the FTDA affected product market entry. Every trademark, upon registration and renewal, must specify one or more USPTO goods-and-services classes in which the trademark is currently being used to sell products. We first compute how many new trademarks are registered, how many expire, and how many active trademarks are in each USPTO class by year. We then use the assignee data to compute

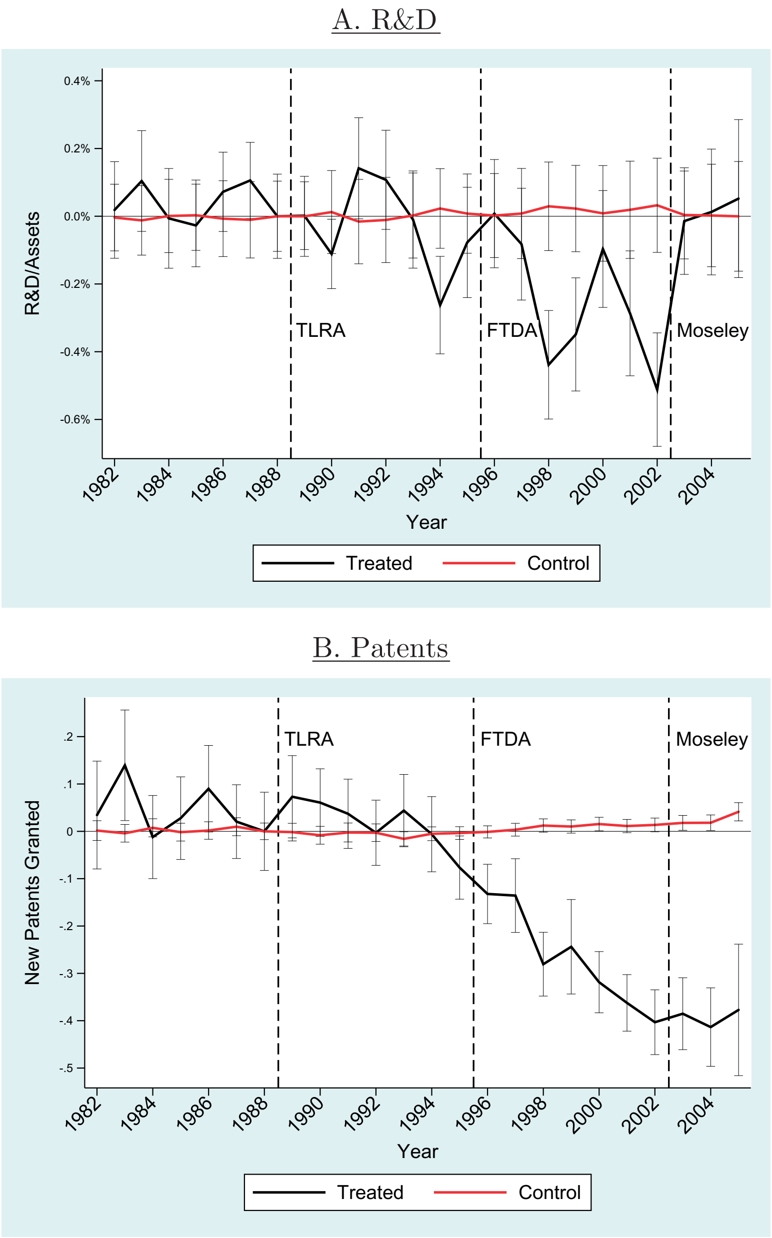


Figure 4
The figure displays the yearly average residual R&D/Assets (panel A) and patents granted (panel B) for the treated and control firm groups after panel regressions with firm and industry-by-year fixed effects. The figure also shows 95% confidence intervals for each group mean in each year. The dashed lines represent the passage of the Trademark Law Revision Act, the passage of the Federal Trademark Dilution Act, and the *Moseley* decision.

the number of active firms (both private and public) and how many firms enter and exit each class by year. Our independent variable $FractionTreated_i$ is the fraction of trademarks in class i as of 1995 that were treated by the FTDA. We standardize $FractionTreated_i$ to have unit variance.

Table 5 shows the FTDA's estimated effects on product market dynamics. Panel A shows that a 1-standard-deviation higher $FractionTreated$ was followed post-FTDA by a 43% relative fall in new trademark registrations, a 14% fall in expiry of older trademarks and a 31% fall in the number of active trademarks. In panel B we see that a 1-standard-deviation higher $FractionTreated$ was followed by a 38% relative fall in new firms entering, an 11% fall in firms exiting and a 28% fall in the number of active firms in that USPTO class. The results are similar when we control for individual pretrends for each class.

Figure 5 shows graphs of the year-by-year coefficients for firm and product entry. The event period spans 1982–2005 as to include the TLRA, FTDA, and Moseley decision. Panel A shows that firm entry was unchanged around the TLRA placebo event, decreased following the FTDA, and returned to pretreatment levels following the *Moseley* decision in early 2003. Panel B shows a similar pattern with new product entry. These results are consistent with our hypotheses that the FTDA changed firms' profits and strategy by increasing entry costs in treated product markets.

The Internet Appendix presents similar estimates using industry-level data from Compustat and the Synthetic Longitudinal Business Database. Industries more affected by the FTDA became more concentrated as measured by their yearly HHI, and new firm listings and new plant openings fell. In sum, the data from all three sources are consistent with our hypothesis that the FTDA raised expected entry costs and led to lower entry and higher concentration in affected product markets and industries.

4.4 Nonlinearity in entry response

Because our results indicate that the FTDA reduced entry in markets with more famous trademarks, next we examine heterogeneity in the strategic response of incumbent firms. A key insight from the strategic investment literature is that entry deterrence motives lead to responses that are nonlinear in the threat of entry faced. Firms that face the highest entry threat have a weak strategic motive, because they face many motivated potential entrants whose best response does not change with the firm's action. Firms that face the lowest entry threat also have a weak strategic motive, because they face only a few weakly motivated potential entrants. Thus, a model of strategic entry deterrence predicts a response that is strongest in firms that face moderate entry threat.[19]

[19] See Dafny (2005), Goolsbee and Syverson (2008), Ellison and Ellison (2011), Frésard and Valta (2016), Cookson (2017a) and Cookson (2017a, 2017b).

Table 5
Effects on entry and exit

A. Trademark registration and expiry

	(1) log $(Register)$	(2) log $(Expire)$	(3) log $(Active)$	(4) log $(Register)$	(5) log $(Expire)$	(6) log $(Active)$
$FracTreated_i$	−0.43**	−0.14***	−0.31***	−0.30**	−0.13***	−0.060**
$\times PostFTDA_t$	(−2.6)	(−6.6)	(−7.5)	(−2.2)	(−6.6)	(−2.1)
Observations	854	854	854	854	854	854
Adjusted R^2	.943	.973	.973	.961	.973	.993
Class FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Pretrend				Quadratic	Quadratic	Quadratic

B. Firm entry and exit

	(1) log $(Entry)$	(2) log $(Exit)$	(3) log $(Firms)$	(4) log $(Entry)$	(5) log $(Exit)$	(6) log $(Firms)$
$FracTreated_i$	−0.38**	−0.11***	−0.28***	−0.25**	−0.11***	−0.063**
$\times PostFTDA_t$	(−2.7)	(−4.3)	(−6.6)	(−2.2)	(−4.3)	(−2.3)
Observations	854	854	854	854	854	854
Adjusted R^2	.933	.968	.975	.955	.968	.994
Class FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Pretrend				Quadratic	Quadratic	Quadratic

The table presents difference-in-differences estimates of the effects of the 1996 Federal Trademark Dilution Act on product market dynamics. Observations are at the USPTO class-by-year level from 1989 to 2002. *FracTreated* is the fraction of trademarks in each trademark class, as of 1995, that were treated by the FTDA, and is standardized to have unit variance. *Register*, *Expired*, and *Active* are the number of new trademarks registered, trademarks that expired, and active trademarks in each class by year. *Entry*, *Exit*, and *Firms* are the number of unique firms that enter, exit, and are active in each class by year. Standard errors are robust and clustered by class.

Table 6 presents estimates of heterogeneity in strategic responses. We interact the treatment coefficient with whether each firm's 1995 Lerner index (industry price-cost margin) was in the high, medium, or low tercile. Industry price-cost margin should be a useful proxy for the threat of entry, because high margins attract potential entrants across a broad class of entry models.[20] Consistent with a model of strategic entry deterrence, the changes in operating profits, product recalls, patenting and new product introductions were all largest in the group that had moderate industry profits ex ante and smaller (though in the same direction) in the groups that had low and high profits. These nonmonotonic patterns in treatment effects depending on the ex ante entry threat are consistent with strategic entry deterrence by the incumbent firms.

4.5 Product market strategy

Finally, Table 7 investigates the FTDA's effects on product market strategy. Columns 1 and 2 show that post-FTDA treated firms reduced their ad spending

20 Ellison and Ellison (2011) use a similar indirect approach to proxy for entry threat via market size.

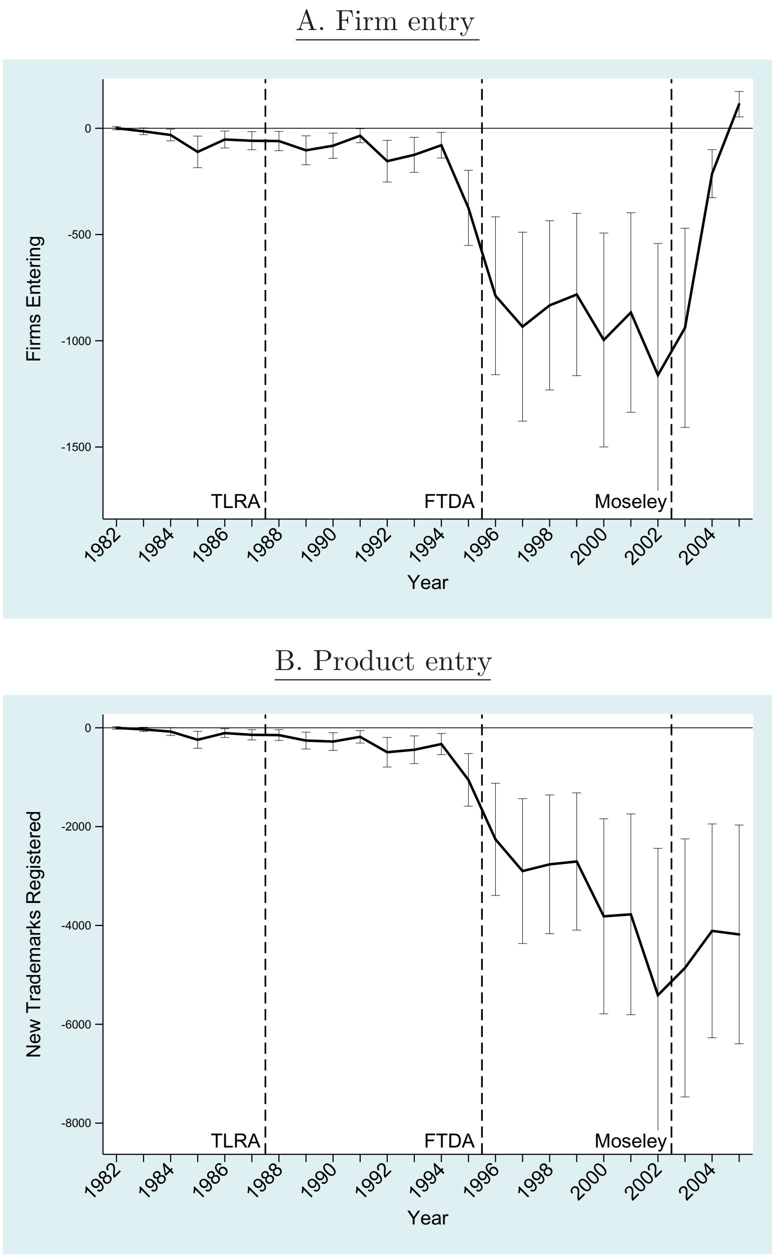


Figure 5
The figure displays the year-by-year coefficient and 95% confidence intervals on $TreatedFraction_i$ as described in the text, where i indexes USPTO goods and service classes. Firm entry is measured from the trademark assignee data (panel A). Product entry is measured from the trademark data (panel B). The dashed lines represent the passage of the Trademark Law Revision Act, the passage of the Federal Trademark Dilution Act, and the *Moseley* decision.

Table 6
Nonlinear responses suggest strategic entry deterrence

	(1) ROA	(2) $hadRecall$	(3) log $(Recalls)$	(4) log $(NewPatents)$	(5) $NewProduct$
$Low1995Lerner_i \times PostFTDA_t \times FamousTM1995_i$	0.018***	0.013*	0.014	−0.08**	−0.13***
	(3.3)	(1.8)	(1.5)	(−2.5)	(−8.6)
$Medium1995Lerner_i \times PostFTDA_t \times FamousTM1995_i$	0.019***	0.016***	0.017***	-0.11**	-0.14***
	(2.8)	(2.7)	(2.8)	(−2.5)	(−8.1)
$High1995Lerner_i \times PostFTDA_t \times FamousTM1995_i$	0.009	0.015	0.016	−0.07	−0.06**
	(1.0)	(1.4)	(1.5)	(−0.7)	(−2.0)
Observations	64,226	64,752	64,752	64,752	64,752
Adjusted R^2	.614	.485	.702	.851	.524
Firm FEs	Yes	Yes	Yes	Yes	Yes
Industry x Year FEs	Yes	Yes	Yes	Yes	Yes

The table presents estimates of heterogeneity in the treatment effects of the 1996 Federal Trademark Dilution Act. Firms are divided into terciles by their industry Lerner index (average price-cost margin) as of 1995. The Lerner index tercile for each firm is then interacted with the difference-in-differences term. *FamousTM1995* is a dummy variable that equals one if the firm held at least one plausibly famous trademark at the end of 1995. *hadRecall* is a dummy variable that equals one if a firm announced at least one recall that year. *Recalls* is the number of recalls by firm-year. *NewPatents* is the number of granted patents the firm applied for in year t. *NewProduct* is a dummy variable that equals one if the firm announced at least one new product that year in the press. Industry-by-year fixed effects are at the 4-digit NAICS level. Standard errors are robust and clustered by firm.

slightly and registered slightly fewer new trademarks per year. By contrast, Columns 3 and 4 show a strong shift in the *type* of trademark registered: treated firms registered 4.2% fewer trademarks in old goods-and-services classes, defined as classes in which the firm ever previously held a trademark, but registered 3.3% more trademarks in new classes, defined as classes in which the firm never previously held a trademark. Column 5 shows that treated firms increased the number of goods-and-services classes in which they were active by 1.4, an increase of 11% relative to their pretreatment average of 13.8 classes. Thus, the FTDA led treated firms to expand their product offerings into new product markets in which they previously had never sold a product.

Table 7, Column 6, shows that treated firms were also more likely to register new trademarks that extended an existing brand, rather than creating a new one. That is, treated firms extended their protected brands into all-new product markets. Finally, Column 7 shows that treated firms were more likely to renew the new trademarks they registered post-FTDA. Thus, treated firms were not only more likely to extend their brands into new markets, but the brand extending products were more likely to still be active in commerce 10 years later, suggesting that the FTDA had long-lived effects on firms' product offerings.

A particularly stark example is Campbell's Inc. The red-and-white Campbell's logo that represents their core brand was first registered in 1932 in a

Table 7
Effects on product market strategy

	(1) $Ad/Sales$	(2) $log(TMs)$	(3) $log(TMs^{Old})$	(4) $log(TMs^{New})$	(5) $Active\ Classes$	(6) $Brand\ Extend$	(7) $TM\ Renew$
$FamousTM1995_i \times PostFTDA_t$	−0.0008 (−1.1)	−0.024 (−1.0)	−0.042* (−1.7)	0.033*** (2.7)	1.42*** (6.4)	0.052*** (4.7)	0.058*** (3.3)
Observations		85,177	86,578	86,578	86,578	19,217	19,217
Adjusted R^2	.682	.671	.706	.131	.898	.789	.251
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry×Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The table presents difference-in-differences estimates of the effects of the 1996 Federal Trademark Dilution Act on firms' product market strategy. *FamousTM1995* is a dummy variable that equals one if the firm held at least one plausibly famous trademark at the end of 1995. *Ad/Sales* is ad expenditure scaled by total sales. *TMs* is the number of new trademarks the firm registered in each year. TMs^{Old} is the number of new trademarks in a USPTO class in which the firm previously held at least one trademark. TMs^{New} is the number of new trademarks in a USPTO class in which the firm never previously held a trademark. *ActiveClasses* is the number of goods and services classes spanned by a firm's active trademarks. *BrandExtend* is the fraction of new trademarks that are extensions of the firm's existing trademarks. *TMRenew* is the fraction of new trademarks that were eventually renewed. Industry-by-year fixed effects are at the 4-digit NAICS level. Standard errors are robust and clustered by firm.

single trademark class, Foods (46). It was renewed in 1952, 1972, and 1993 and was still active in the same one class in 1995, 63 years after its initial registration. In 1996 the firm registered the Campbell's logo in fifteen new trademark classes including Cutlery (25), Crockery and Porcelain (30), Glassware (33), Paper and Stationery (37), Clothing (39), Jewelry (28), Furniture (32), and Unclassified (50). Considering that each of these new classes was subject to the use-in-commerce requirement,[21] those new registrations represented a significant broadening of Campbell's product market offerings in precisely the year the FTDA became effective.

Figure 6 shows graphs of the year by year coefficients for new product announcements and registrations of products in all-new goods and service classes. Panel A shows that treated and control firms' new product announcements trended similarly around the TLRA placebo event, fell after the passage of the FTDA, and leveled off following the *Moseley* decision. Panel B shows that the fraction of trademarks registered in new product classes was unchanged around the TLRA placebo event, and increased after the passage of the FTDA and the *Moseley* decision. These year-by-year coefficient estimates are consistent with the estimates in Table 7 and with our hypotheses that the following the FTDA, treated firms moved from an innovative toward an exploitative product market strategy.

21 The Internet Appendix shows examples of Campbell's branded furniture and jewelry that were produced and sold in the late 1990s.

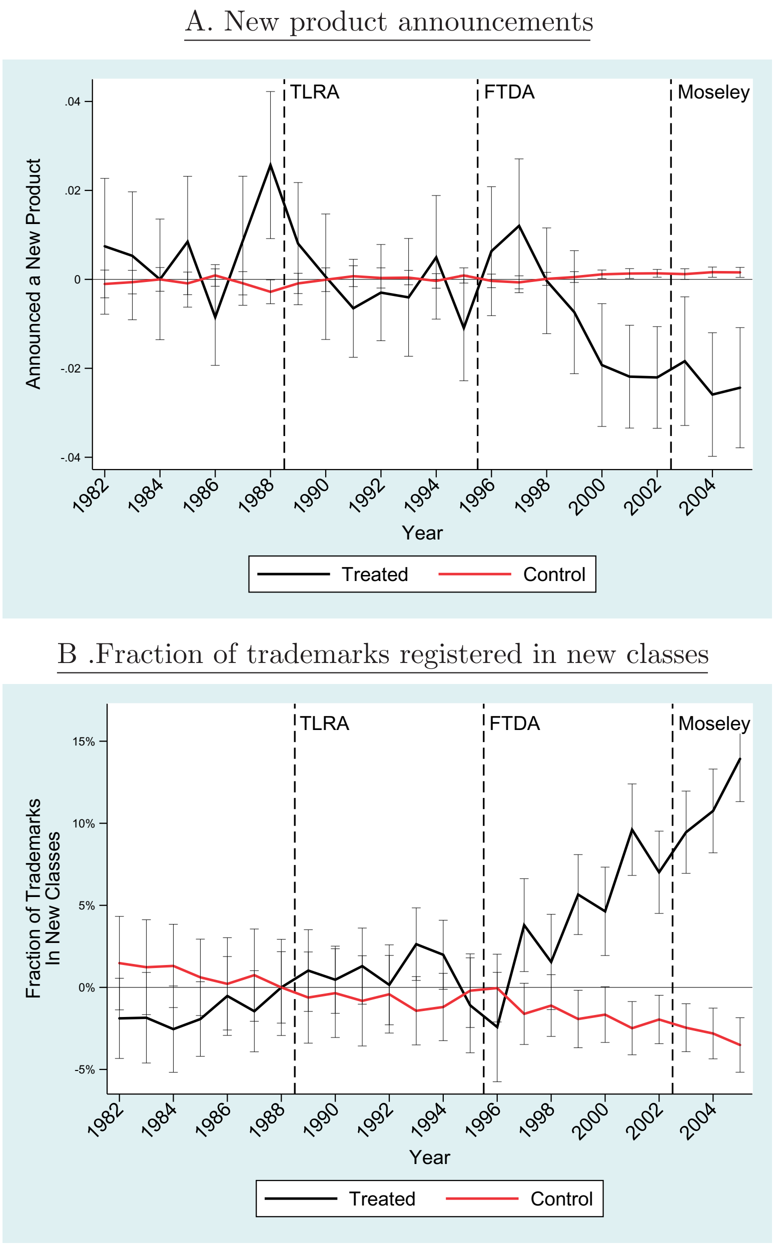


Figure 6
The figure displays the yearly average residual of new products announced in the *New York Times* (panel A) and the fraction of trademarks registered in new product classes (panel B) for the treated and control firm groups after panel regressions with firm and industry-by-year fixed effects. The figure also shows 95% confidence intervals for each group mean in each year. The dashed lines represent the passage of the Trademark Law Revision Act, the passage of the Federal Trademark Dilution Act, and the *Moseley* decision.

5. Conclusion

In surveys conducted by the NSF and Census Bureau, both private and public firms rate trademarks as their most important intellectual property assets, ahead of trade secrets, copyrights, and patents (Jankowski 2012). This paper uses the 1996 Federal Trademark Dilution Act to examine the effects of trademark protection on firms' profits and strategy. We document that treated firms' operating profits increased sharply after the FTDA's passage, and fell back after its key provision was nullified by a Supreme Court decision. The value of added trademark protection was localized in industries that had high advertising expenditures, selling expenditures, and trademark holdings ex ante. More affected product markets had sharply lower entry and exit post-FTDA, consistent with a mechanism whereby the FTDA raised entry costs for potential entrants into affected product markets. Treatment effects were strongest in firms that faced moderate entry threat ex ante, consistent with strategic entry deterrence by the incumbent firms.

Guided by theory, we investigate the effects of strengthening trademark protection on firm strategy. Treated firms reduced product quality as measured by increased incidence and value of product recalls, and curtailed innovation by reducing R&D spending, patenting, and new product announcements. While these observations are broadly consistent with a quiet life or rent extraction hypothesis, we also find that treated firms introduced new products into previously unexplored markets by extending their brands into all-new goods and service classes. Taken together, these observations suggest that stronger trademark protection led firms to pursue exploitative rather than exploratory innovation and product market strategies.

It is important to acknowledge that our results show the effects of the treatment on the treated, that is, the effects of granting additional antidilution protection to incumbent firms in a strong IP enforcement regime. This episode is useful to understand how strengthening the protection of famous trademarks affected firm and industry dynamics. An open question is whether these effects dominate in the long run, and whether there are benefits to strengthening trademark protection for smaller firms and startups or in environments of weak IP enforcement. The comprehensive USPTO trademarks data can contribute to further research on these topics, as well as new research on product market dynamics and intangible capital.

We believe these results are an important first step, by demonstrating that firms' profits and strategy are strongly affected by changes in trademark protection at the margin. Our findings that the FTDA was followed by lower product quality and innovation add to recent studies on the potential downside of intellectual property protection (Boldrin and Levine 2002; Williams 2013; Moser 2013), and suggest that strengthening trademark protection may not deliver what is promised on the label.

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